

A density-two sheared hyperbolic-secant Gabor nonframe

Automated mathematics run `fa_banach_001`

Lane 8, `agent_lane_08`, model GPT5.6

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Status. Candidate counterexample, likely valid, with a directed-interval Poincaré–Miranda certificate. Human verification is recommended.

Abstract

Faulhuber and Shafkulovska ask for the lattice frame set of the Fourier-invariant hyperbolic secant and note that sufficient density conditions are unknown for general lattices. We give a computer-assisted existence proof of a non-separable lattice of density 2 on which the normalized hyperbolic secant does not generate a Gabor frame. The mechanism is a simultaneous pair of Zak-transform zeros after a metaplectic shear. One zero is forced by evenness; the other is certified by the Poincaré–Miranda theorem with explicit interval and tail bounds.

1 Source question and scope

The source is Faulhuber and Shafkulovska, *Gabor frame bound optimizations*, arXiv:2204.02917 [1]. On page 13, after solving the rectangular optimization problem for the hyperbolic secant, they write:

“In contrast to the Gaussian case, we do not know sufficient density conditions for the hyperbolic secant and general lattices to yield Gabor frames. So, before speculating about an optimality result for general lattices one should work out the lattice frame set of the hyperbolic secant.”

$$h_B\left(\frac{\eta}{n}\right) = h_B\left(\frac{1}{\eta}\right) \iff \frac{\eta}{n} = \frac{1}{\eta} \iff \eta = \sqrt{n}.$$

Furthermore, for all $\eta > \sqrt{n}$ we have

$$\frac{d}{d\eta} B\left(\frac{\eta}{n}, \frac{1}{\eta}\right) > 0.$$

Due to the symmetry about $\sqrt{n}$, the square lattice $n^{-1/2}\mathbb{Z} \times n^{-1/2}\mathbb{Z}$ is the unique minimizing lattice of B among rectangular lattices of density $n \in \mathbb{N}$.

As A is maximized by the square lattice $n^{-1/2}\mathbb{Z} \times n^{-1/2}\mathbb{Z}$ and B is minimized by the square lattice $n^{-1/2}\mathbb{Z} \times n^{-1/2}\mathbb{Z}$, also the condition number $\kappa = B/A$ has to be minimal in this case. In contrast to the Gaussian case, we do not know sufficient density conditions for the hyperbolic secant and general lattices to yield Gabor frames. So, before speculating about an optimality result for general lattices one should work out the lattice frame set of the hyperbolic secant. Nonetheless, we note that the conjecture of Strohmer and Beaver [49] was based on the radial symmetry of the Gaussian in the time-frequency plane and connections to sphere packings (see also [4]). By the characterization of radially symmetric ambiguity functions, which can only come from Gaussians and Hermite functions [21], it is clear that the hyperbolic secant does not possess a radial symmetry in the time-frequency plane.

Figure 1: The page-13 passage in the original paper. The crop retains the full readable page width and the complete surrounding paragraph.

The theorem below disproves the most direct extension of the rectangular criterion: density greater than one is not sufficient for arbitrary lattices. It does not determine the full lattice frame set, and it does not exclude sufficient conditions that use lattice shape in addition to density.

2 Definitions

For $x, \omega \in \mathbb{R}$, let

$$T_x f(t) = f(t - x), \quad M_\omega f(t) = e^{2\pi i \omega t} f(t), \quad \pi(x, \omega) = M_\omega T_x.$$

For a full-rank lattice $\Lambda \subset \mathbb{R}^2$, write $\mathcal{G}(f, \Lambda) = \{\pi(\lambda)f : \lambda \in \Lambda\}$. The lattice density is the reciprocal of its covolume.

For $a > 0$, use the Zak transform

$$Z_a f(x, \omega) = \sum_{k \in \mathbb{Z}} f(x - ak) e^{2\pi i ak \omega}.$$

All windows used below are Schwartz functions, so their Zak transforms are continuous and all displayed series may be differentiated termwise on the compact parameter boxes in question.

Finally, let

$$g(t) = \sqrt{\frac{\pi}{2}} \operatorname{sech}(\pi t), \quad C_q f(t) = e^{\pi i q t^2} f(t).$$

The scalar normalization of g is immaterial for zeros, but agrees with the source paper.

3 Proof intuition

At density two, the Zak representation has two fibers. A frame requires that the sum of their squared moduli stay uniformly positive. For an even window, the two fibers automatically share one zero at the half-time point and half-frequency shift. It therefore suffices to tune a chirp until the other fiber vanishes at the same half-time point.

The relevant Zak value is a two-parameter complex series $F(a, c)$. A high-precision search locates a zero extremely close to $(1/\sqrt{28}, 15/28)$. The formal proof does not assert that this center is an exact zero. Instead, it preconditions $(\Re F, \Im F)$, proves the correct signs on the four faces of a 10^{-6} -radius box, and invokes Poincaré–Miranda. A chirp/shear covariance then moves the nonframe from a chirped window on a rectangular lattice to the original hyperbolic secant on a sheared lattice of the same density.

4 The density-two Zak obstruction

Lemma 1 (Two-fiber obstruction). *Let f have continuous Zak transform, $a > 0$, and $b = 1/(2a)$. If, for some (x_0, ω_0) ,*

$$Z_a f(x_0, \omega_0) = Z_a f(x_0, \omega_0 + b) = 0,$$

then $\mathcal{G}(f, a\mathbb{Z} \times b\mathbb{Z})$ is not a frame.

Proof. On the Zak rectangle $Q = [0, a) \times [0, 1/a)$, direct index changes give

$$\begin{aligned} Z_a(T_{na}f)(x, \omega) &= e^{-2\pi i na \omega} Z_a f(x, \omega), \\ Z_a(M_{mb}f)(x, \omega) &= e^{2\pi i mb x} Z_a f(x, \omega - mb). \end{aligned}$$

Write $m = 2\ell + r$, $r \in \{0, 1\}$. Since $2b = 1/a$ is the frequency period of $Z_a f$, the two residue classes of m give the two multipliers

$$Z_a f(x, \omega), \quad e^{\pi i x/a} Z_a f(x, \omega - b),$$

times the orthogonal Fourier basis $e^{2\pi i \ell x/a} e^{-2\pi i n a \omega}$ on Q . Parseval therefore shows that a lower Gabor frame bound would force

$$|Z_a f(x, \omega)|^2 + |Z_a f(x, \omega - b)|^2 \geq A' > 0$$

for almost every $(x, \omega) \in Q$, with an inessential normalization constant A' . The two asserted zeros, quasi-periodicity, and continuity make the left side zero at (x_0, ω_0) and arbitrarily small on positive-measure neighborhoods. No positive essential lower bound exists. $\square$

5 Certified existence of the tuned chirp

Define, for $a > 0$ and $c \in \mathbb{R}$,

$$F(a, c) = \sum_{k \in \mathbb{Z}} \operatorname{sech}(\pi a(k + \tfrac{1}{2})) e^{\pi i c(k + \frac{1}{2})^2}. \quad (1)$$

Set

$$a_0 = \frac{1}{\sqrt{28}}, \quad c_0 = \frac{15}{28}, \quad r = 10^{-6},$$

and let

$$B = \{(a, c) : |a - a_0| \leq r, |c - c_0| \leq r\}.$$

Lemma 2 (Certified zero box). *There is $(a_*, c_*) \in B$ such that $F(a_*, c_*) = 0$.*

Proof. Put $U = (\Re F, \Im F)^T$, $J_0 = DU(a_0, c_0)$, and use the fixed decimal-rational preconditioner

$$A = \begin{pmatrix} 0.07386071204 & -0.11010607132 \\ -0.01062523295 & -0.02565158148 \end{pmatrix}, \quad G = AU.$$

Directed interval summation, including explicit tails, gives

$$\begin{aligned} |\Re F(a_0, c_0)|, |\Im F(a_0, c_0)| &< 10^{-40}, \\ F_a(a_0, c_0) &\in [8.37043247980345, 8.37043247980346] \\ &\quad + i[-3.46714665606285, -3.46714665606284], \\ F_c(a_0, c_0) &\in [-35.928990826924, -35.928990826923] \\ &\quad + i[-24.101675899640, -24.101675899639]. \end{aligned}$$

In particular, every entry of $AJ_0 - I_2$ has absolute value below 10^{-9} .

For completeness, the uniform analytic remainder bound is short. If $x = k + 1/2$, termwise differentiation of (1) gives, for either real component,

$$|F_{aa}| \leq \pi^2 \sum_x |x|^2 \operatorname{sech}(\pi a|x|), \quad |F_{ac}| \leq \pi^2 \sum_x |x|^3 \operatorname{sech}(\pi a|x|), \quad |F_{cc}| \leq \pi^2 \sum_x |x|^4 \operatorname{sech}(\pi a|x|).$$

Throughout B , $a > 0.188$. Using $\operatorname{sech} y \leq 2e^{-y}$, $\pi^2 < 10$, and $e^{-0.59} < 0.555$, the elementary differentiated-geometric-series bounds give

$$|F_{aa}| < 400, \quad |F_{ac}| < 2100, \quad |F_{cc}| < 14000.$$

Consequently, the Taylor remainder in either real component of U is at most

$$\frac{1}{2}(400 + 2 \cdot 2100 + 14000)r^2 = 9.3 \times 10^{-9}.$$

The absolute row sums of A are 0.18396678336 and 0.03627681443. Thus the two transformed remainders are at most

$$1.710892 \times 10^{-9}, \quad 3.37375 \times 10^{-10}.$$

Combining the center, linear, and remainder bounds yields

$$\begin{aligned} |G_1(a, c) - (a - a_0)| &< 1.711 \times 10^{-9} + 2 \times 10^{-15}, \\ |G_2(a, c) - (c - c_0)| &< 3.374 \times 10^{-10} + 2 \times 10^{-15} \end{aligned}$$

for $(a, c) \in B$. Hence G_1 is strictly negative and positive on the two a -faces, respectively, while G_2 is strictly negative and positive on the two c -faces. Poincaré–Miranda gives a point in B where $G = 0$. Since A is invertible, $F = 0$ there.

The finite directed-interval calculation and the differentiated geometric tail estimates are reproduced by `code/verify_zero_box.py`. $\square$

6 Counterexample theorem

Theorem 3. *There exist a_*, c_* satisfying*

$$\left| a_* - \frac{1}{\sqrt{28}} \right| \leq 10^{-6}, \quad \left| c_* - \frac{15}{28} \right| \leq 10^{-6},$$

such that, with $q_ = c_*/a_*^2$, the lattice*

$$\Lambda_* = \mathbb{Z}(a_*, -q_*a_*) + \mathbb{Z}\left(0, \frac{1}{2a_*}\right) \quad (2)$$

has density 2, but $\mathcal{G}(g, \Lambda_)$ is not a frame. The lattice is not a coordinate-product lattice.*

Proof. Choose (a_*, c_*) from Lemma 2, put $q_* = c_*/a_*^2$, $b_* = 1/(2a_*)$, and $h = C_{q_*}g$. At the half-time point,

$$Z_{a_*}h(a_*/2, 0) = \sqrt{\frac{\pi}{2}} F(a_*, c_*) = 0.$$

The window h is even. At the half-frequency shift,

$$Z_{a_*}h(a_*/2, b_*) = \sum_{k \in \mathbb{Z}} h(a_*(\frac{1}{2} - k))(-1)^k = 0,$$

because the terms indexed by k and $1 - k$ have opposite signs and equal window values. Lemma 1 says that $\mathcal{G}(h, a_*\mathbb{Z} \times b_*\mathbb{Z})$ is not a frame.

It remains only to undo the chirp. Direct calculation gives

$$C_q^{-1}\pi(x, \omega)C_q = e^{\pi i q x^2} \pi(x, \omega - qx).$$

Unitary conjugation and irrelevant unimodular atom phases therefore identify $\mathcal{G}(C_q g, a\mathbb{Z} \times b\mathbb{Z})$ with

$$\mathcal{G}(g, \mathbb{Z}(a, -qa) + \mathbb{Z}(0, b)).$$

This gives exactly (2). Its covolume is $a_*b_* = 1/2$, so its density is 2.

Finally, a coordinate-product lattice would require the vertical shear of the first generator to be an integral multiple of b_* , i.e. $q_*a_*/b_* = 2c_* \in \mathbb{Z}$. But the certified box puts $2c_*$ strictly between 1 and 2. Thus the lattice is genuinely non-separable in the given time-frequency coordinates. $\square$

7 Verification and limitations

The verifier was run with

```
conda run --no-capture-output -n sandbox python \  
  runs/fa_banach_001/solutions/counterexamples/\  
  2204.02917_density_two_sheared_sech_nonframe/code/verify_zero_box.py
```

It printed **CERTIFIED**. In particular, the strict face margins were larger than 9.98×10^{-7} , while the largest nonlinear transformed remainder was below 1.72×10^{-9} . The proof uses the interval library only for finite evaluations of elementary functions; convergence, tails, and Taylor remainders are bounded analytically.

The result is existential within a tiny explicit box. It does not claim that the visually distinguished center $(1/\sqrt{28}, 15/28)$ is itself an exact zero. It supplies one density-two nonframe, not a classification of all shears or all densities.

8 Bounded novelty check

On 21 July 2026 we searched the run registry and arXiv using the exact source id and combinations of **hyperbolic secant**, **general lattice**, **sheared lattice**, **non-frame**, and **lattice frame set**. The rectangular theorem of Janssen and Strohmer [2], the half-irregular product-set theorem of Baranov and Belov [3], and the later Hermite-window paper [4] were checked as close variants. None contains the general-lattice sech counterexample above. This is a bounded, arXiv-centered search, so novelty confidence is moderate rather than definitive.

9 Human-review recommendation

We recommend high-priority expert review. The three shortest checks are: (i) rerun the interval certificate in an independent ball-arithmetic system, (ii) inspect Lemma 1's two-fiber normalization, and (iii) verify the sign in the last chirp covariance. The substantial mathematical claim is otherwise reduced to explicit inequalities with large margins.

References

- [1] M. Faulhuber and I. Shafkulovska, *Gabor frame bound optimizations*, arXiv:2204.02917 (2022; revised 2023).
- [2] A. J. E. M. Janssen and T. Strohmer, *Hyperbolic secants yield Gabor frames*, arXiv:math/0301134 (2003).
- [3] A. Baranov and Y. Belov, *Irregular sampling for hyperbolic secant type functions*, arXiv:2312.10174 (2023).
- [4] M. Faulhuber, I. Shafkulovska, and I. Zlotnikov, *On the frame property of Hermite functions and analytic extensions of their frame sets*, arXiv:2403.10503 (2024).
- [5] C. Miranda, *Un'osservazione su un teorema di Brouwer*, Bollettino dell'Unione Matematica Italiana 3 (1940), 5–7.